

Paper

A study on latent structural models for binary relational data with attribute information

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Abstract: This study focuses on relational data obtained through object relations. Traditional analysis of relational data often ignores attribute information. Therefore, Mikawa *et al.* proposed a method to estimate the latent structure of continuous relational data using a generative model and parameter estimation. However, real-world relational data can be discrete, and therefore, we propose a new model for binary relational data using a generative model based on the Bernoulli distribution and the Monte Carlo Expectation-Maximization (EM) algorithm for parameter estimation. We also clarify the effectiveness of the proposed model through simulation experiments using artificial data and real data.

Key Words: statistical relational learning, latent structural model, Monte Carlo EM algorithm, Laplace approximation

1. Introduction

Various types of data have become available due to the recent development of information technology. Due to this movement, the analysis data with relationships between data (relational data) is also attracting attention. Typical examples of relational data are the information about human relationships, customer and product purchases, and ratings. When considering relational data, there is a method, called statistical relational learning, which expresses the relationships between objects and extracts the relationships between the data [1]. Statistical relational learning is used for social



network analysis [2] such as community detection [3], customer purchase or rating prediction, link prediction [4], and so on. Many statistical relational learning methods only focus on the relationships between the data [5, 6]. As mentioned before, due to the development of information technology, attribute information of each object is stored in relational information. However, there are few statistical relational learning methods that focus on the attribute information. Using this information for relational data analysis, one can build an analytical model considering the relationship between objects.

To efficiently utilize attribute information, Mikawa *et. al.* have proposed a latent structural model of relational data and how to estimate the latent structure of the data [7]. The conventional model assumes that its relations are represented in continuous values and defines a generative model for relational data with attribute information. In addition, the model proposes a way to estimate the latent structure and parameter estimation. Furthermore, due to the difficulty of calculating the Q function which is often used for estimating latent structure, the Monte Carlo EM algorithm is used for approximating the Q function. Even though the model above works well when relations are represented in continuous values, the relations can be given as discrete values when dealing with actual data (*e.g.* whether hyperlinks exist between web pages, how many items a user purchased). The model assumes continuous values for relations between objects, however, it cannot be applied to discrete values.

From the above discussion, in this study, we define a generative model for binary relational data and propose a latent structure analysis model assuming that relations are represented in discrete values, and provide a method for estimating latent structure and parameters. Specifically, we assume that the relationship is given in binary (0 or 1) following the Bernoulli distribution, and perform parameter estimation using the Monte Carlo EM algorithm. We also execute simulation experiments and clarify the effectiveness of the proposed model using artificial data and real data.

This paper is organized as follows. In Sect. 2, our related studies, definitions of variables for our study and the Monte Carlo EM Algorithm which is one of the parameter estimation methods are introduced. In addition, the conventional model [7], which is a latent structural model of continuous relational data with attribution information and how to estimate the latent structure of the data, is explained. In Sect. 3, we introduce the proposed model. In Sect. 4 and Sect. 5, in order to clarify the effectiveness of the proposed model, we conduct simulation experiments using artificial data and real data, respectively. We further discuss the simulation results. In Sect. 6, conclusions and future works are presented.

2. Preliminaries

2.1 Statistical relational learning

Relational data [8] is data that expresses the relationship between objects. Single domain and multiple domain are typical types of relational data, in which data is represented as a relationship between homogeneous objects (*e.g.* user relations in social media, the hyperlink between websites) and a relationship between heterogeneous objects (*e.g.* relationships between item and user in purchase history data), respectively [9, 10]. Based on the relational data, the study on relational data analysis has been done (*e.g.* link prediction such as item recommendation, information propagation in a network [11, 12], and knowledge discovery such as community detection [13]).

In general, relational data has only relationships between objects, which does not have any attribute information about each object. Although various analytical methods have been proposed [14, 15], these approaches generally deal with such relational data that does not have attribute information. On the other hand, in this study, we assume that features of each object for relational data are given. We define a generative model under the assumption of relational data and propose a parameter estimation method for the binary relational data.

2.2 Notations and assumptions

Let $\mathbf{x}_i = (x_{i1}, x_{i2}, \dots, x_{id})^\top \in \mathbb{R}^d, 1 \leq i \leq n$, be the i -th observed data (object) of dimension d . The *observed variable matrix* $\mathbf{X} = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n)$ is the $d \times n$ matrix whose i -th column corresponds to

$\mathbf{x}_i$.

For each $\mathbf{x}_i$, we assume that it is generated from the latent variable $\mathbf{z}_i = (z_{i1}, z_{i2}, \dots, z_{ik})^\top \in \mathbb{R}^k$ of dimension k , where $k < d$ and $\mathbf{z}_i \sim \mathcal{N}(\mathbf{0}, \sigma_z^2 I)$. The *latent variable matrix* $\mathbf{Z} = (\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_n)$ is the $k \times n$ matrix whose i -th column corresponds to $\mathbf{z}_i$. We denote by $\mathbf{Z}_{\setminus i}$ the $k \times (n-1)$ matrix obtained from $\mathbf{Z}$ by deleting the i -th column.

Let $F \in \mathbb{R}^{d \times k}$ be a projection matrix that maps $\mathbf{z}_i$ to $\mathbf{x}_i$ with the mean term $\boldsymbol{\mu}_x$ and the noise term $\boldsymbol{\varepsilon}_i$. Herein, we assume that $\mathbf{x}_i$ is generated by

$$\mathbf{x}_i = F\mathbf{z}_i + \boldsymbol{\mu}_x + \boldsymbol{\varepsilon}_i, \quad (1)$$

where $\boldsymbol{\varepsilon}_i \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$ and $\boldsymbol{\Sigma} = \text{diag}(\sigma_1^2, \sigma_2^2, \dots, \sigma_d^2)$ is a $d \times d$ diagonal matrix whose diagonal entries are σ_m^2 ($m = 1, 2, \dots, d$). We also consider the relation between distinct inputs $\mathbf{x}_i$ and $\mathbf{x}_j$, and denote by $\mathbf{Y} = (y_{ij})$ the $n \times n$ matrix whose (i, j) -entry y_{ij} corresponds to the relation between $\mathbf{x}_i$ and $\mathbf{x}_j$. Throughout this paper, we assume that $y_{ij} \in \{0, 1\}$ and y_{ij} follows the Bernoulli distribution $\text{Bern}(s_{ij})$ such that $p(y_{ij} = 1) = s_{ij}$ for some $0 \leq s_{ij} \leq 1$. By convention, we always set $y_{ij} = 1$ if $i = j$.

To summarize, for the input data $\mathbf{x}_i$, the latent variable $\mathbf{z}_i$ and the relation y_{ij} between $\mathbf{x}_i$ and $\mathbf{x}_j$, we have that

$$\begin{aligned} \mathbf{x}_i &\sim \mathcal{N}(F\mathbf{z}_i + \boldsymbol{\mu}_x, \boldsymbol{\Sigma}); \\ \mathbf{z}_i &\sim \mathcal{N}(\mathbf{0}, \sigma_z^2 I); \text{ and} \\ y_{ij} &\sim \text{Bern}(s_{ij}), \end{aligned}$$

where we set s_{ij} to be

$$s_{ij} = \frac{1}{1 + \exp\{-w_0 - w\mathbf{z}_i^\top \mathbf{z}_j\}}, \quad (2)$$

for some fixed w_0 and w .

2.3 Monte Carlo EM algorithm

An EM algorithm [16], consisting of the E (Expectation)-step and the M (Maximization)-step, is one of the methods to find maximum likelihood estimators under statistic models. An EM algorithm generally implements a parameter estimation using latent class models. However, there exist some models under which the update equation in the EM algorithm, specifically the E-step, may not be analytically feasible. For such models, the Monte Carlo EM algorithm can be used to approximate the E-step based on sampling [17].

Let $\boldsymbol{\theta}^{old}$ and $\boldsymbol{\theta}$ be the parameter vectors before and after the update, respectively. In the original EM algorithm, the Q function is then defined as follows using the latent variable matrix $\mathbf{Z}$ and the observed variable matrix $\mathbf{X}$.

$$Q(\boldsymbol{\theta}, \boldsymbol{\theta}^{old}) = \int p(\mathbf{Z}|\mathbf{X}, \boldsymbol{\theta}^{old}) \ln p(\mathbf{Z}, \mathbf{X}|\boldsymbol{\theta}) d\mathbf{Z}. \quad (3)$$

By contrast, the Monte Carlo EM algorithm approximates the integral of the right-hand side of Eq. (3) using a finite set $\{\mathbf{Z}^{(l)}\}$ consisting of L samples $\mathbf{Z}^{(1)}, \mathbf{Z}^{(2)}, \dots, \mathbf{Z}^{(L)}$ of latent variable matrices generated from the current estimate $p(\mathbf{Z}|\mathbf{X}, \boldsymbol{\theta}^{old})$ as follows.

$$Q(\boldsymbol{\theta}, \boldsymbol{\theta}^{old}) \simeq \frac{1}{L} \sum_{l=1}^L \ln p(\mathbf{Z}^{(l)}, \mathbf{X}|\boldsymbol{\theta}). \quad (4)$$

As for the M-step, on the other hand, the same way as in the original EM algorithm is used to maximize the Q function.

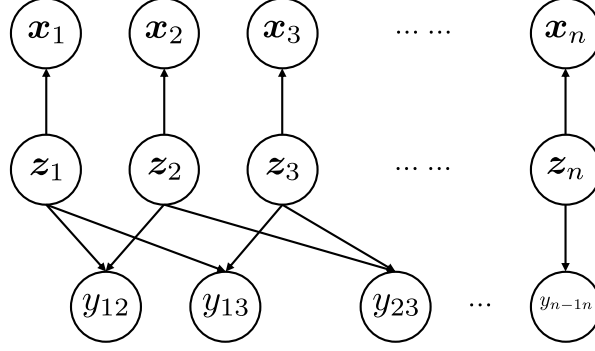


Fig. 1. Graphical model of the conventional model.

2.4 Latent structural models for continuous relational data [7]

Mikawa *et. al.* have proposed a latent structural model of continuous relational data with attribution information and how to estimate the latent structure of the data [7]. The conventional model assumes that the relation y_{ij} between the i -th and the j -th objects is generated from

$$y_{ij} = \mathbf{z}_i^\top \mathbf{z}_j + e_{ij}, \quad (5)$$

where e_{ij} follows $e_{ij} \sim \mathcal{N}(0, \sigma_y^2)$. Figure 1 illustrates the graphical model of the conventional model.

From Fig. 1, the model assumes that the input vector $\mathbf{x}_i$ is generated by the latent vector $\mathbf{z}_i$ under Eq. (1). In addition, the relation y_{ij} is generated by latent vectors $\mathbf{z}_i, \mathbf{z}_j$ under the Eq. (5).

Let $\boldsymbol{\theta}$ be the parameter vector whose elements are $F, \boldsymbol{\mu}_x, \boldsymbol{\Sigma}, \sigma_z^2, \sigma_y^2$. From Fig. 1, the likelihood function $\mathcal{L}$ for $\boldsymbol{\theta}$ is given by

$$\mathcal{L} = \prod_{i=1}^n p(\mathbf{x}_i | F, \mathbf{z}_i, \boldsymbol{\mu}_x, \boldsymbol{\Sigma}) \left\{ \prod_{i \neq j}^n p(y_{ij} | \mathbf{z}_i, \mathbf{z}_j, \sigma_y^2) \right\}^{\frac{1}{2}}. \quad (6)$$

In order to apply the Monte Carlo EM algorithm to the model, the posterior distribution $p(\mathbf{Z}^{(l)}, \mathbf{X}, \mathbf{Y} | \boldsymbol{\theta})$ for the latent variable $\mathbf{Z}$ is essential. Herein, the Q function is approximated by the following equation

$$\begin{aligned} Q(\boldsymbol{\theta}, \boldsymbol{\theta}^{old}) &\simeq \frac{1}{L} \sum_{l=1}^L \ln p(\mathbf{Z}^{(l)}, \mathbf{X}, \mathbf{Y} | \boldsymbol{\theta}) \\ &= \frac{1}{L} \sum_{l=1}^L \ln \left\{ p(\mathbf{Z}^{(l)} | \boldsymbol{\theta}) p(\mathbf{X} | \mathbf{Z}^{(l)}, \boldsymbol{\theta}) p(\mathbf{Y} | \mathbf{Z}^{(l)}, \boldsymbol{\theta}) \right\} \\ &= \frac{1}{L} \sum_{l=1}^L \left\{ \ln p(\mathbf{Z}^{(l)} | \boldsymbol{\theta}) + \ln p(\mathbf{X} | \mathbf{Z}^{(l)}, \boldsymbol{\theta}) + \ln p(\mathbf{Y} | \mathbf{Z}^{(l)}, \boldsymbol{\theta}) \right\}, \end{aligned} \quad (7)$$

where $\mathbf{Z}^{(1)}, \mathbf{Z}^{(2)}, \dots, \mathbf{Z}^{(L)}$ are generated from the posterior distribution of $\mathbf{Z}$.

For the parameter estimation, the model uses the Monte Carlo EM algorithm. However, it needs to estimate the posterior distribution of $\mathbf{Z}$. The posterior distribution $p(\mathbf{z}_i | \mathbf{x}_i, \mathbf{Y}, \mathbf{Z}_{\setminus i}, \boldsymbol{\theta})$ is obtained as $p(\mathbf{z}_i | \mathbf{x}_i, \mathbf{Y}, \mathbf{Z}_{\setminus i}, \boldsymbol{\theta}) \sim \mathcal{N}(\boldsymbol{\nu}_i, M_i)$, and $\boldsymbol{\nu}_i, M_i$ are derived as follows:

$$M_i = \left\{ \frac{1}{\sigma_z^2} I + \frac{1}{2\sigma_y^2} \sum_{j \neq i} \mathbf{z}_j \mathbf{z}_j^\top + F^\top \boldsymbol{\Sigma}^{-1} F \right\}^{-1}. \quad (8)$$

$$\boldsymbol{\nu}_i = A_i^{-1} \left(F^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i + \boldsymbol{\mu}_x) + \frac{1}{\sigma_y^2} \sum_{j \neq i} y_{ij} \mathbf{z}_j \right). \quad (9)$$

In addition, parameters $\boldsymbol{\theta}$, which maximize Eq. (7), can be given by the following equations

Algorithm 1: Parameter estimation for the conventional model

```

1 Initialize parameter vector  $\theta$ .
2 while  $\theta$  does not converge do
3   for  $l = 1, 2, \dots, L$  do
4     for  $i = 1, 2, \dots, n$  do
5       Calculate  $\nu_i$  and  $M_i$ .
6       Generate a sample  $\mathbf{z}_i$  by  $p(\mathbf{z}_i | \mathbf{x}_i, \mathbf{Y}, \mathbf{Z}_{\setminus i}, \theta) \sim \mathcal{N}(\nu_i, M_i)$ .
7        $\mathbf{z}_i^{(l)} \leftarrow \mathbf{z}_i$ .
8     end
9     Obtain  $\mathbf{Z}^{(l)} = (\mathbf{z}_1^{(l)}, \mathbf{z}_2^{(l)}, \dots, \mathbf{z}_n^{(l)})$ .
10  end
11  Using  $\mathbf{Z}^{(1)}, \mathbf{Z}^{(2)}, \dots, \mathbf{Z}^{(L)}$ , compute parameters  $F, \mu_x, \Sigma, \sigma_y^2$  and  $\sigma_z^2$  based on
    Eqs. (10)–(14).
12 end

```

$$F = \left(\sum_{l=1}^L \sum_{i=1}^n (\mathbf{x}_i - \mu_x) \mathbf{z}_i^{(l)\top} \right) \left(\sum_{l=1}^L \sum_{i=1}^n \mathbf{z}_i^{(l)} \mathbf{z}_i^{(l)\top} \right)^{-1}; \quad (10)$$

$$\mu_x = \frac{1}{Ln} \sum_{l=1}^L \sum_{i=1}^n (\mathbf{x}_i - F \mathbf{z}_i^{(l)}); \quad (11)$$

$$\Sigma = \text{diag} \left(\frac{1}{Ln} \sum_{l=1}^L \sum_{i=1}^n (\mathbf{x}_i - F \mathbf{z}_i^{(l)} - \mu_x) (\mathbf{x}_i - F \mathbf{z}_i^{(l)} - \mu_x)^\top \right); \quad (12)$$

$$\sigma_y^2 = \frac{1}{Lnh} \sum_{l=1}^L \sum_{i=1}^n \sum_{j \neq i}^n \left(y_{ij} - (\mathbf{z}_i^{(l)})^\top \mathbf{z}_j^{(l)} \right)^2; \quad (13)$$

$$\sigma_z^2 = \frac{1}{Lkn} \sum_{l=1}^L \sum_{i=1}^n \left\| \mathbf{z}_i^{(l)} \right\|^2; \quad (14)$$

where $\mathbf{z}_i^{(l)}$ is the i -th column of $\mathbf{Z}^{(l)}$ and $h = \binom{n}{2}$, the number of choosing distinct i and j amongst n choices. The conventional model performs parameter estimation by mutually repeating two steps which are: (1) calculating the posterior distribution of $\mathbf{z}_i$ and approximating the Q function using the distribution; and (2) updating parameters by using the approximated Q function. Here, the computation of the posterior distribution of $\mathbf{z}_i$ and the parameter estimation can be implemented by **Algorithm 1**.

2.5 Laplace approximation

In our proposed model, we calculate the posterior distribution of $\mathbf{z}_i$ and estimate the parameters using the Monte Carlo EM algorithm with samples generated from the posterior distribution. However, since the probability distribution of y_{ij} follows a Bernoulli distribution, we cannot derive the exact posterior distribution like [7]. Therefore, we use Laplace approximation [17] to approximate the posterior distribution of $\mathbf{z}_i$.

The Laplace approximation for the function $f(\mathbf{r})$ approximates the probability density $p(\mathbf{r})$ assumed in the form of Eq. (15) by a normal distribution $\mathcal{N}(\mathbf{r}_0, \mathbf{C}^{-1})$ whose mean is the maximum point $\mathbf{r}_0$ and covariance matrix is $\mathbf{C}^{-1}$.

$$p(\mathbf{r}) = \frac{f(\mathbf{r})}{\int f(\mathbf{r}) d\mathbf{r}}. \quad (15)$$

Here, $\mathbf{C}$ is given by $\mathbf{C} = -\nabla \nabla \ln f(\mathbf{r})|_{\mathbf{r}=\mathbf{r}_0}$, where ∇ means the gradient operator.

3. Proposed model

In this section, we will explain the proposed model. More precisely, we propose a latent structural model assuming that the relationship is given in binary (0 or 1). Figure 1 also illustrates the graphical model of our proposed model. That is, there is no difference in the graphical models, however, we strongly emphasize that there is a huge difference in which y_{ij} is a continuous value or binary value. Due to the difference, the conventional model cannot be directly applied to the relational data represented in binary values. In other words, it is necessary to derive the posterior distribution of $\mathbf{z}_i$ to approximate the Q function, however, it is difficult to calculate the posterior distribution of $\mathbf{z}_i$ in the closed form. Herein, we explain an approach to address such problems and a parameter estimation method in our proposed model.

Let $\boldsymbol{\theta}$ be the parameter vector whose elements are $F, \boldsymbol{\mu}_x, \boldsymbol{\Sigma}, w_0, w, \sigma_z^2$. From Fig. 1, the likelihood function $\mathcal{L}$ for $\boldsymbol{\theta}$ is given by

$$\mathcal{L} = \prod_{i=1}^n p(\mathbf{x}_i | F, \mathbf{z}_i, \boldsymbol{\mu}_x, \boldsymbol{\Sigma}) \left\{ \prod_{i \neq j}^n p(y_{ij} | \mathbf{z}_i, \mathbf{z}_j, w_0, w) \right\}^{\frac{1}{2}}. \quad (16)$$

In the case of using the EM algorithm, the expected value of the log-likelihood as the Q function is calculated by the following equation.

$$Q(\boldsymbol{\theta}, \boldsymbol{\theta}^{old}) = \int p(\mathbf{Z} | \mathbf{X}, \mathbf{Y}, \boldsymbol{\theta}^{old}) \ln p(\mathbf{Z}, \mathbf{X}, \mathbf{Y} | \boldsymbol{\theta}) d\mathbf{Z}. \quad (17)$$

In addition, parameter estimation is done by optimizing $\boldsymbol{\theta}$. In this study, we assume that $y_{ij} \sim \text{Bern}(s_{ij})$, but it is difficult to compute the integral part within the Q function when applying the EM algorithm under the condition. We, therefore, apply the Monte Carlo EM algorithm described above for the parameter estimation. Here, the Q function of our proposed model is approximated by

$$\begin{aligned} Q(\boldsymbol{\theta}, \boldsymbol{\theta}^{old}) &\simeq \frac{1}{L} \sum_{l=1}^L \ln p(\mathbf{Z}^{(l)}, \mathbf{X}, \mathbf{Y} | \boldsymbol{\theta}) \\ &= \frac{1}{L} \sum_{l=1}^L \ln \{ p(\mathbf{Z}^{(l)} | \boldsymbol{\theta}) p(\mathbf{X} | \mathbf{Z}^{(l)}, \boldsymbol{\theta}) p(\mathbf{Y} | \mathbf{Z}^{(l)}, \boldsymbol{\theta}) \} \\ &= \frac{1}{L} \sum_{l=1}^L \{ \ln p(\mathbf{Z}^{(l)} | \boldsymbol{\theta}) + \ln p(\mathbf{X} | \mathbf{Z}^{(l)}, \boldsymbol{\theta}) + \ln p(\mathbf{Y} | \mathbf{Z}^{(l)}, \boldsymbol{\theta}) \}, \end{aligned} \quad (18)$$

where $\mathbf{Z}^{(1)}, \mathbf{Z}^{(2)}, \dots, \mathbf{Z}^{(L)}$ are generated from the posterior distribution of $\mathbf{Z}$. To implement the Monte Carlo EM algorithm, we need to calculate the posterior distribution $p(\mathbf{Z} | \mathbf{X}, \mathbf{Y}, \boldsymbol{\theta})$ for the latent variable matrix $\mathbf{Z}$, and generate samples that follow the distribution.

In performing the Monte Carlo EM algorithm, however, it is difficult to compute the posterior distribution of $\mathbf{Z}$ in the closed form. As such, we use Laplace approximation to approximate the posterior distribution of $\mathbf{z}_i$ to $\mathcal{N}(\boldsymbol{\eta}_i, A_i)$. Since we can generate data samples from $\mathcal{N}(\boldsymbol{\eta}_i, A_i)$, the Q function is also able to be approximated. Therefore, it is possible to optimize parameter $\boldsymbol{\theta}$ using the approximated Q function.

3.1 The posterior distribution for the latent variable

As mentioned above, to apply the Monte Carlo EM algorithm, it is necessary to obtain $p(\mathbf{Z} | \mathbf{X}, \mathbf{Y}, \boldsymbol{\theta})$ for the latent variable matrix $\mathbf{Z}$, and generate samples that follow the distribution. The posterior distribution $p(\mathbf{Z} | \mathbf{X}, \mathbf{Y}, \boldsymbol{\theta})$ for $\mathbf{Z}$ is defined as

$$\begin{aligned} p(\mathbf{Z} | \mathbf{X}, \mathbf{Y}, \boldsymbol{\theta}) &= \frac{p(\mathbf{X}, \mathbf{Y}, \mathbf{Z} | \boldsymbol{\theta})}{\int p(\mathbf{X}, \mathbf{Y}, \mathbf{Z} | \boldsymbol{\theta}) d\mathbf{Z}} \\ &= \frac{p(\mathbf{Z} | \boldsymbol{\theta}) p(\mathbf{X} | \mathbf{Z}, \boldsymbol{\theta}) p(\mathbf{Y} | \mathbf{Z}, \boldsymbol{\theta})}{\int p(\mathbf{X}, \mathbf{Y}, \mathbf{Z} | \boldsymbol{\theta}) d\mathbf{Z}} \end{aligned}$$

$$= \frac{\prod_{i=1}^n p(\mathbf{z}_i, \sigma_z^2 I) p(\mathbf{x}_i | \mathbf{z}_i, F, \boldsymbol{\mu}_x, \boldsymbol{\Sigma}) \prod_{j \neq i}^n p(y_{ij} | \mathbf{z}_i, \mathbf{z}_j, w_0, w)^{\frac{1}{2}}}{\int \prod_{i=1}^n p(\mathbf{z}_i, \sigma_z^2 I) p(\mathbf{x}_i | \mathbf{z}_i, F, \boldsymbol{\mu}_x, \boldsymbol{\Sigma}) \prod_{j \neq i}^n p(y_{ij} | \mathbf{z}_i, \mathbf{z}_j, w_0, w)^{\frac{1}{2}} d\mathbf{z}_i}. \quad (19)$$

It is impossible to exactly calculate the posterior distribution. For this reason, we apply the Laplace approximation in order to obtain an approximation of the posterior distribution.

Herein, $\ln f(\mathbf{Z} | \mathbf{X}, \mathbf{Y}, \boldsymbol{\theta})$ is defined as the logarithm of numerator of Eq. (19), which is given by the following equation.

$$\begin{aligned} \ln f(\mathbf{Z} | \mathbf{X}, \mathbf{Y}, \boldsymbol{\theta}) = & \sum_{i=1}^n \left(\ln \frac{1}{(2\pi)^{\frac{d}{2}} |\sigma_z^2 I|^{\frac{1}{2}}} - \frac{1}{2\sigma_z^2} \|\mathbf{z}_i\|^2 + \ln \frac{1}{(2\pi)^{\frac{d}{2}} |\boldsymbol{\Sigma}|^{\frac{1}{2}}} \right. \\ & - \frac{1}{2} (\mathbf{x}_i - F\mathbf{z}_i - \boldsymbol{\mu}_x)^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - F\mathbf{z}_i - \boldsymbol{\mu}_x) \\ & \left. + \frac{1}{2} \sum_{j \neq i} \{y_{ij} \ln s_{ij} + (1 - y_{ij}) \ln(1 - s_{ij})\} \right). \end{aligned} \quad (20)$$

In contrast, the posterior distribution of $\mathbf{z}_i$, where mean is $\boldsymbol{\eta}_i$ and covariance matrix is A_i , is approximated using Laplace approximation. The approximation derives that

$$p(\mathbf{z}_i | \mathbf{x}_i, y_{ij}, \mathbf{Z}_{\setminus i}, \boldsymbol{\theta}) \sim \mathcal{N}(\boldsymbol{\eta}_i, A_i), \quad (21)$$

where A_i is the second-order differential matrix of the negative log-likelihood and is given by

$$A_i = \left(\frac{1}{\sigma_z^2} I + F^\top \boldsymbol{\Sigma}^{-1} F + \frac{1}{2} \sum_{j \neq i} s_{ij} (1 - s_{ij}) w^2 \mathbf{z}_j \mathbf{z}_j^\top \right)^{-1}, \quad (22)$$

and $\boldsymbol{\eta}_i$ is obtained by maximizing the numerator of Eq. (20).

As for $\boldsymbol{\eta}_i$, it is difficult to directly obtain an optimal solution, which also implies that A_i cannot be directly obtained, either. Therefore, we adopt a gradient method to solve $\boldsymbol{\eta}_i$. In this study, we use the Newton method [18] for estimating $\boldsymbol{\eta}_i$. Now, differentiating Eq. (20) by $\mathbf{z}_i$ is given by the following equation.

$$\frac{\partial \ln f(\mathbf{Z} | \mathbf{X}, \mathbf{Y}, \boldsymbol{\theta})}{\partial \mathbf{z}_i} = -\frac{1}{\sigma_z^2} \mathbf{z}_i + F \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - F\mathbf{z}_i - \boldsymbol{\mu}_x) + \frac{1}{2} \sum_{j \neq i} (y_{ij} - s_{ij}) w \mathbf{z}_j. \quad (23)$$

See the Appendix A for the derivation of the A_i and $\boldsymbol{\eta}_i$.

As mentioned before, after deriving the posterior distribution of $\mathbf{z}_i$, the Q function is approximated by generated sample $\mathbf{Z}^{(l)}$ using the Eq. (18). As in the Monte Carlo EM algorithm, each parameter can be updated to maximize the Q function. In other words, the updating formula of each parameter can be derived by differentiating the approximated Q function with each parameter.

3.2 Estimation on w and w_0

Same as the conventional model, parameters of $F, \boldsymbol{\mu}_x, \boldsymbol{\Sigma}, \sigma_z^2$ can be updated using Eqs. (10)–(12) and (14) for each parameter after approximating the Q function. That is, the updating formula can be derived as maximizing the Eq. (18). See the appendix B for derivation of parameters $F, \boldsymbol{\mu}_x, \boldsymbol{\Sigma}, \sigma_z^2$.

From Eqs. (10)–(13), while those parameters can be directly computed, parameters w_0 and w are not obtained as a closed form. We instead calculate their gradients and update them using a gradient-based method.

Here, the gradients of w_0 and w are computed as follows:

$$\frac{\partial}{\partial w_0} \frac{1}{L} \sum_{l=1}^L \ln p(\mathbf{Z}^{(l)}, \mathbf{X}, \mathbf{Y} | \boldsymbol{\theta}) = \frac{1}{L} \sum_{l=1}^L \sum_{i=1}^n \sum_{j \neq i}^n (y_{ij} - s_{ij}), \quad (24)$$

$$\frac{\partial}{\partial w} \frac{1}{L} \sum_{l=1}^L \ln p(\mathbf{Z}^{(l)}, \mathbf{X}, \mathbf{Y} | \boldsymbol{\theta}) = \frac{1}{L} \sum_{l=1}^L \sum_{i=1}^n \sum_{j \neq i}^n (y_{ij} - s_{ij}) \mathbf{z}_i^\top \mathbf{z}_j. \quad (25)$$

Algorithm 2: Parameter estimation for the proposed model

```
1 Initialize parameter vector  $\theta$ .
2 while  $\theta$  does not converge do
3   for  $l = 1, 2, \dots, L$  do
4     for  $i = 1, 2, \dots, n$  do
5       Calculate  $\eta_i$  and  $A_i$ .
6       Generate a sample  $z_i$  by  $p(z_i|x_i, Y, Z_{\setminus i}, \theta) \sim \mathcal{N}(\eta_i, A_i)$ .
7        $z_i^{(l)} \leftarrow z_i$ .
8     end
9     Obtain  $Z^{(l)} = (z_1^{(l)}, z_2^{(l)}, \dots, z_n^{(l)})$ .
10  end
11  Using  $Z^{(1)}, Z^{(2)}, \dots, Z^{(L)}$ , compute parameters  $F, \mu_x, \Sigma$  and  $\sigma_z^2$  based on Eqs. (10)–(12)
    and (14), and compute gradients of  $w_0$  and  $w$  based on Eqs. (24) and (25).
12 end
```

In this study, we use the Adam [19] for updating w_0 and w .

From the above discussions, the computation of the posterior distribution of z_i and the parameter estimation can be implemented by following **Algorithm 2**.

4. Simulation experiment using artificial data

In order to clarify the effectiveness of the proposed model, we conducted simulation experiments using artificial data.

4.1 Simulation setting

The simulation experiments are the followings of: (1) confirming the accuracy of parameter estimation when k is varied; (2) whether the number of dimension for a latent vector can be decided using the BIC [20]; and (3) confirming a performance by changing the number of data n and the dimension d of input data.

The experimental setting is as follows;

Table I. Simulation settings.

	dimension d of data	# of data n	# of samples L	# of iterations	# of trials	latent dimension k	true latent dimension k^*
Experiment 1	15	150	10	10	10	from 1 to 6	3
Experiment 2	15	150	10	10	10	from 1 to 10	1, 3, 5, 7, 9
Experiment 3	varied	varied	10	10	10	3	3

Here, based on the preliminary experiments, we assume that both the number of samples and the number of iterations are set to 10 since those numbers do not significantly affect the experimental results. In addition, we use the Newton method [18] for estimating η_i which is a mean of the posterior distribution of z_i , and the Adam [19] for estimating w_0 and w which are parameters of y_{ij} . We execute 10 trials to implement the algorithm with different initial values, and obtain the values of the RMSE and Q function. Additionally, we calculate the mean value of the RMSE and the Q function for each iteration over 10 trials. Mathematically speaking, we first obtain the value $S_{v,r}$ of the r -th iteration in the v -th trial. We then calculate the mean value of each iteration as $\bar{S}_{(r)} = \frac{1}{10} \sum_{v=1}^{10} S_{v,r}$ for Experiments 1, 3, and 4. In addition, we also determine the best value $\min_r \bar{S}_{\min(r)}$ across all trials and iterations, where $S_{\min(r)} = \min_v S_{v,r}$.

Since the proposed model is a generative model of relational data, it can generate artificial data. We use the following procedures to generate artificial data, where $*$ indicates the true parameter.

Step 1) Set $\sigma_z^{2*} = 1$ and generate $Z^* = (z_1^*, z_2^*, \dots, z_n^*)$ so that $z_i^* \sim \mathcal{N}(\mathbf{0}, \sigma_z^{2*} I)$.

Step 2) Set the random values for w_0^* and w^* , and compute s_{ij}^* based on Eq. (2). Generate $\mathbf{Y}^* = (y_{ij}^*)$ so that $y_{ij}^* \sim \text{Bern}(s_{ij}^*)$.

Step 3) Generate $\mathbf{X} = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n)$ so that $\mathbf{x}_i \sim \mathcal{N}(F^* \mathbf{z}_i^* + \boldsymbol{\mu}_x^*, \boldsymbol{\Sigma}^*)$, where each entry of F^* follows $\mathcal{N}(0, 10)$, $\boldsymbol{\Sigma}^* = \text{diag}(0.1, 0.1, \dots, 0.1)$ and $\boldsymbol{\mu}_x^* = \mathbf{0}$.

Experiments 1 and 2 are done using artificial data generated by fixing $n = 150, d = 15$. On the other hand, Experiment 3 is performed by varying input data conditions.

4.2 Experiment 1; Parameter estimations when the dimension of the latent vector is varied

In this subsection, we consider the performance of parameter estimations when the dimension k of the latent vector is varied. Note that since our proposed model is a generative model, we can set the true dimension k^* , and we verify whether the proposed model can grasp the true dimension. In this experiment, we set the true dimension $k^* = 3$ and the latent vector dimension k ranges from 1 to 6, and calculate the RMSE each parameters and value of Q function.

We show the results on Experiment 1 in Fig. 2. The horizontal axis of Fig. 2 shows the number of iterations, and the vertical axis shows the RMSE between true value and estimated one or the value of Q function. Figure 2 emphasizes the importance of analyzing the appropriate dimension k in advance. Indeed, estimations of parameters work the best when the k derived from matches. From Fig. 2, when the latent vector dimension k is smaller than the true dimension k^* , the performance of parameter estimation is relatively low in most cases. On the other hand, in the case of $k \geq 3$, the RMSEs of y_{ij} and F are almost the same. From the result of the RMSE of y_{ij} and the RMSEs of w_0 and w , we observe that the parameter estimation performance of w_0, w hardly affects the estimation of y_{ij} .

Table II shows the best results in this experiment. In this case, the tendency of the result is similar to Fig. 2. From the above discussions, a more expressive model is constructed when $k > k^* = 3$, while the performance of the RMSE is not drastically changed. Therefore, it is important to find the true dimension k^* to analyze latent structure, so we verify whether the Bayesian Information Criterion (explained in the next subsection) can decide an appropriate dimension in Experiment 2.

4.3 Experiment 2; Identifying the true dimension of the latent structure

Generally, the empirical decision, such as interpretability of output, is adapted when deciding the number of latent dimensions in the latent class model. Using this, there is a possibility that different results may be derived when the analyst changes. In this experiment, we consider whether the Bayesian Information Criterion (BIC) [20], which is calculated as

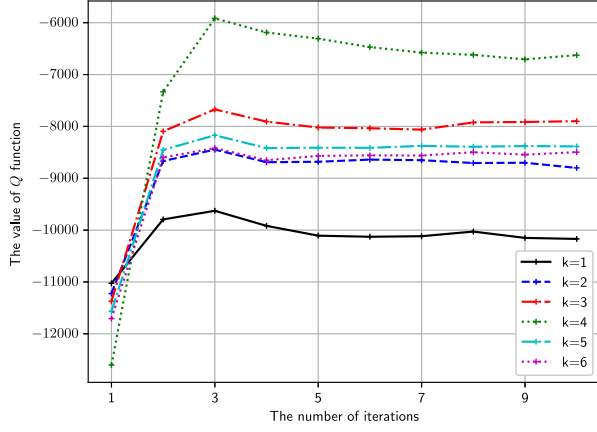
$$BIC = -2 \ln \mathcal{L} + \left((k+1)d - \frac{k(k-1)}{2} \right) \ln n, \quad (26)$$

can be used to derive the true dimension of the latent vector or not. If so, then we can avoid the empirical decisions explained above, which results in reducing the computational complexity. Note that the true dimension k^* affects the likelihood function of $\mathcal{L}$ from Eq. (16). The reasons of using the BIC as the model selection criterion are that 1) the objective of this study is to estimate the latent structure underlying the observed data, and that 2) BIC possesses consistency in this regard.

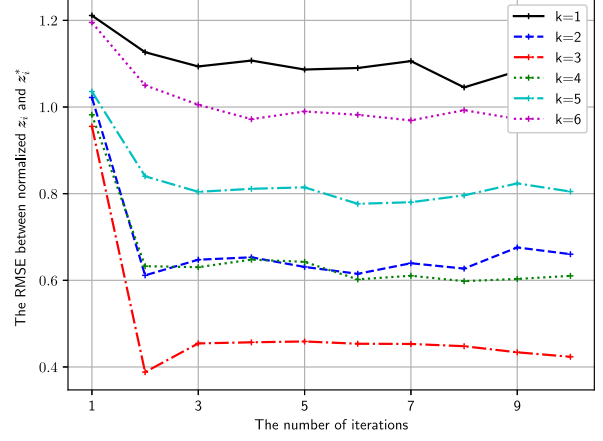
In this experiment, the true dimension k^* of the latent vector is set to be 1, 3, 5, 7, 9, and we calculate the BIC when k ranges from 1 to 10 for each k^* . If the best (smallest) BIC is given when $k = k^*$, then it is suitable to use the BIC for deciding the dimension of the latent vector.

Figure 3 shows the result of Experiment 2. In Fig. 3, the horizontal and vertical axes denote the number of dimensions and the value of the BIC, respectively.

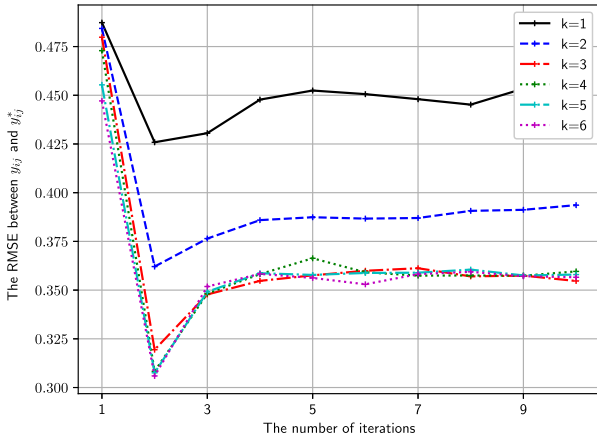
From Fig. 3, we can observe that the BIC is smallest when $k = k^*$ for each k^* , and therefore, the BIC is appropriate for deciding k^* .



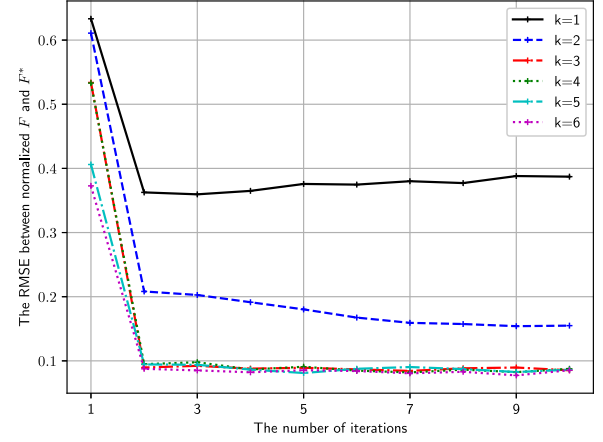
(a) The value of the Q function



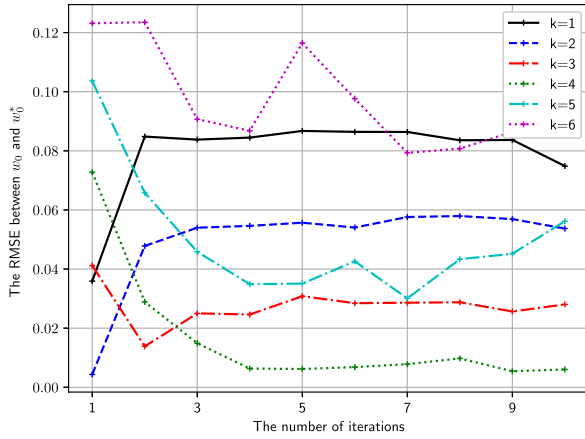
(b) The RMSE between z_i and z_i^*



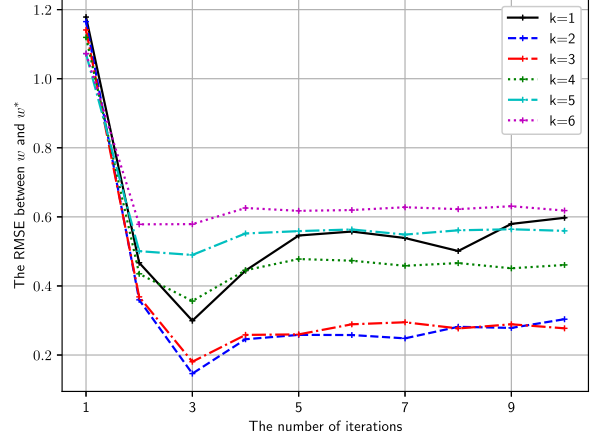
(c) The RMSE between y_{ij} and y_{ij}^*



(d) The RMSE between F and F^*



(e) The RMSE between w_0 and w_0^*



(f) The RMSE between w and w^*

Fig. 2. Experimental results when k is varied.

4.4 Experiment 3; Parameter estimations when input data condition is varied

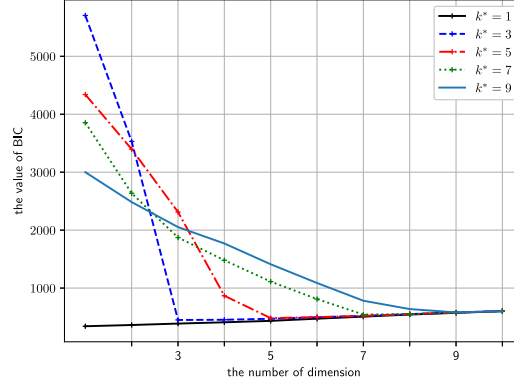
Under the condition that latent vector dimension k can be decided using the BIC, we confirm the performance of parameter estimation when the condition of input data n, d is varied.

We conduct the following two cases of simulation, where the first case is varying n and the second case is varying d . The simulation setting of input data is as follows:

1. $L = 10, d = 15, n = 50, 100, 150, 200, 250, 300$

Table II. The smallest RMSEs for each k (Experiment 1).

	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$	$k = 6$
Σ	0.5270	0.2260	0.0802	0.0805	0.0815	0.0744
$\mathbf{Z}$	1.0356	0.5710	0.3337	0.5647	0.7489	0.9363
F	0.3271	0.1368	0.0687	0.0652	0.0717	0.0676
$\mathbf{Y}$	0.4146	0.3542	0.3170	0.3013	0.2981	0.2930
$\mathbf{X}$	0.7452	0.5365	0.4020	0.4007	0.4154	0.4105
w_0	0.0286	0.0012	0.0118	0.0002	0.0223	0.0585
w	0.1945	0.1069	0.1455	0.3165	0.4656	0.5076

**Fig. 3.** The BIC when k^* is varied.**Table III.** The smallest RMSEs for each n (Experiment 3).

	$n = 50$	$n = 100$	$n = 150$	$n = 200$	$n = 250$	$n = 300$
Σ	0.1538	0.1080	0.0698	0.0643	0.0657	0.0483
$\mathbf{Z}$	0.4361	0.4270	0.2921	0.2867	0.2922	0.2476
F	0.1062	0.0597	0.0599	0.0489	0.0373	0.0361
$\mathbf{Y}$	0.2175	0.2423	0.2353	0.2389	0.2343	0.2246
$\mathbf{X}$	0.4635	0.4483	0.3645	0.3754	0.3787	0.3496
w_0	0.0084	0.0389	0.0144	0.1011	0.1434	0.1618
w	0.8744	0.7011	0.6533	0.5598	0.5450	0.5179

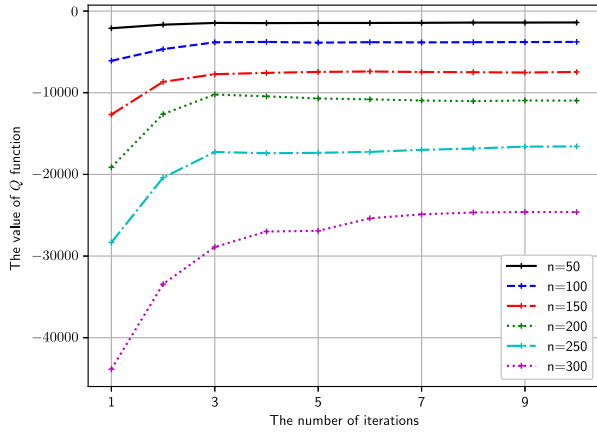
Table IV. The smallest RMSEs for each d (Experiment 3).

	$d = 5$	$d = 10$	$d = 15$	$d = 20$	$d = 25$	$d = 30$
Σ	0.2153	0.1171	0.0932	0.0979	0.0689	0.0617
$\mathbf{Z}$	0.7378	0.4532	0.4053	0.3839	0.3280	0.3102
F	0.1227	0.0764	0.0738	0.0788	0.0506	0.0632
$\mathbf{Y}$	0.2739	0.2159	0.2152	0.2115	0.2069	0.2038
$\mathbf{X}$	0.5593	0.4377	0.4160	0.4316	0.4053	0.3916
w_0	0.1759	0.1811	0.1800	0.1830	0.1802	0.1785
w	1.8413	1.5068	1.4515	1.4090	1.272	1.234

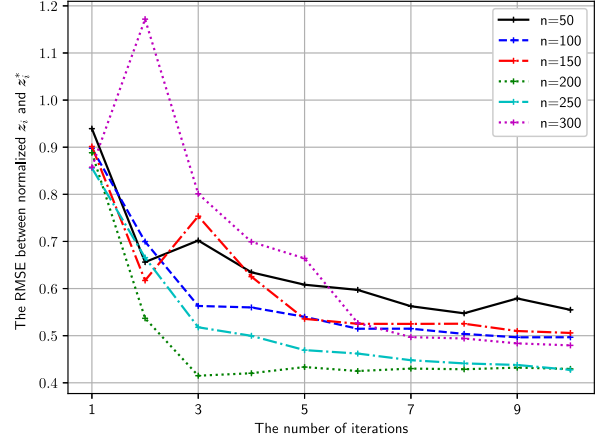
2. $L = 10, d = 5, 10, 15, 20, 25, 30, n = 150$

We verify the performance of the RMSE and the value of Q function, as in Experiment 1. The performance of parameter estimation is shown by Figs. 4 and 5, and those best results are shown by Tables III and IV.

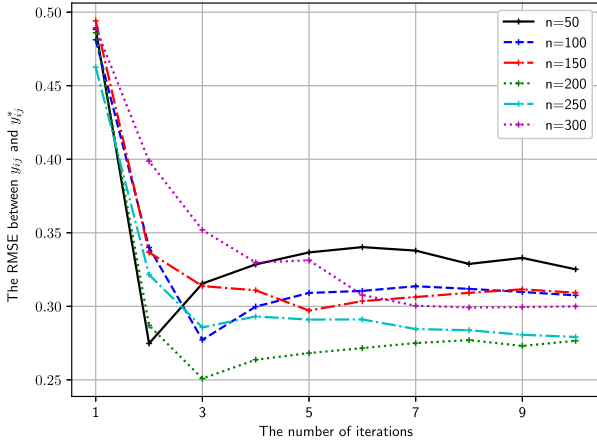
From Fig. 4, the performance of parameter estimation improves as n increases in most cases. In this simulation setting, it especially seems that the parameter estimation is enough performance when the number of data $n = 100$ from the Fig. 4.



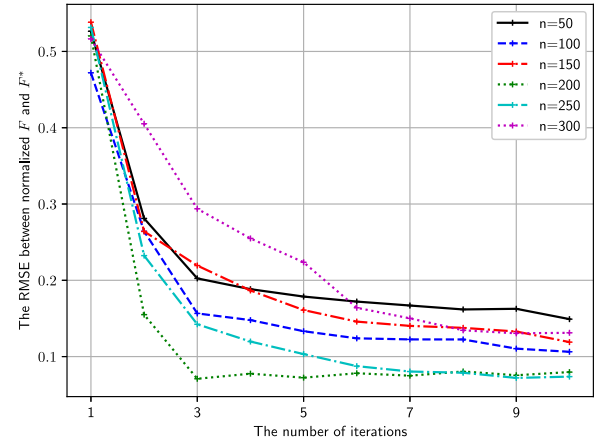
(a) The value of the Q function



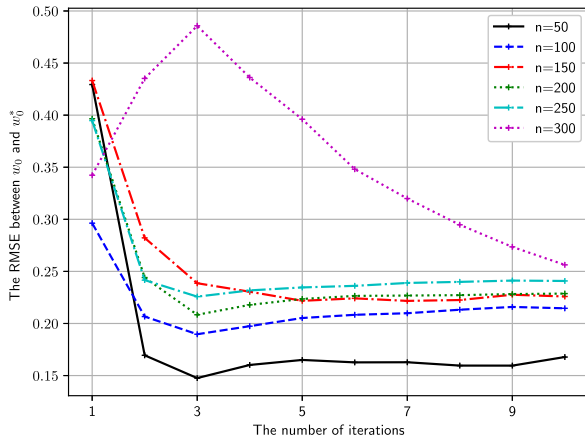
(b) The RMSE between z_i and z_i^*



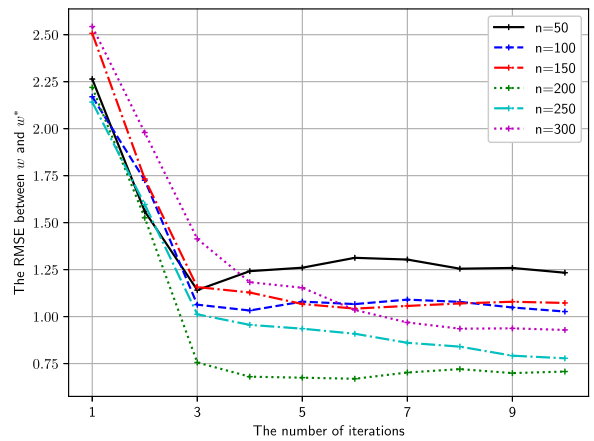
(c) The RMSE between y_{ij} and y_{ij}^*



(d) The RMSE between F and F^*



(e) The RMSE between w_0 and w_0^*

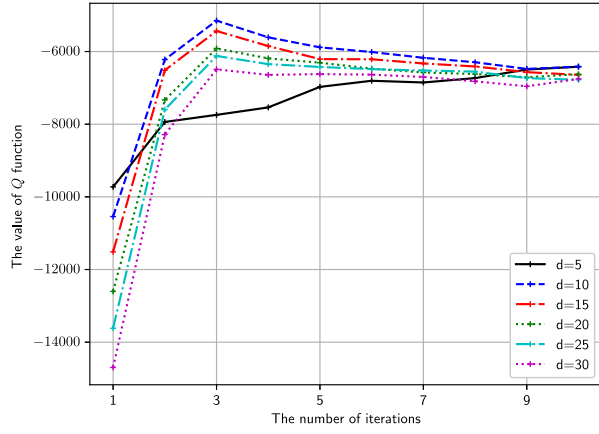


(f) The RMSE between w and w^*

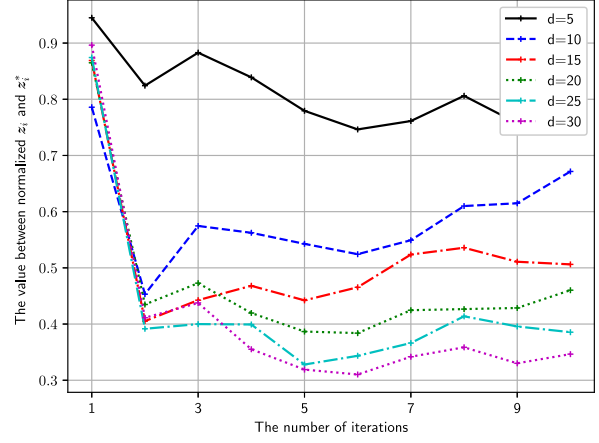
Fig. 4. Experimental results when n is varied.

Since the number of parameters to be estimated for $\mathbf{Z}$ is $n \times k$ and that for $\mathbf{Y}$ is $\frac{n(n-1)}{2}$, those numbers seem to be not large enough for estimation, and therefore, the performance of parameter estimation might be considered to be relatively low in the case of $n = 50$.

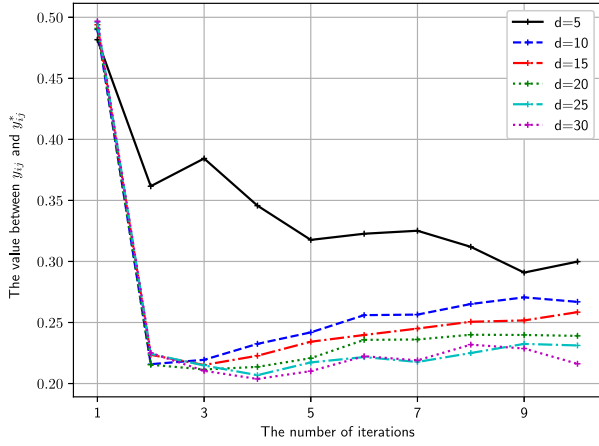
On the other hand, from the Fig. 5, the performance of parameter estimation is improved when the number of dimension d increases. It seems that due to increasing the dimension d of the input data while keeping the true latent dimension is fixed at $k^* = 3$, the amount of information for parameter



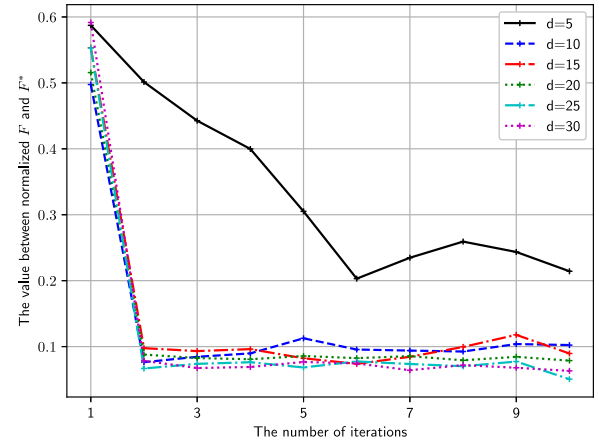
(a) The value of the Q function



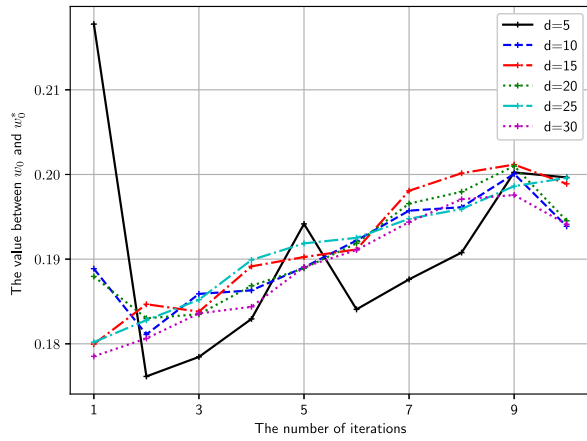
(b) The RMSE between z_i and z_i^*



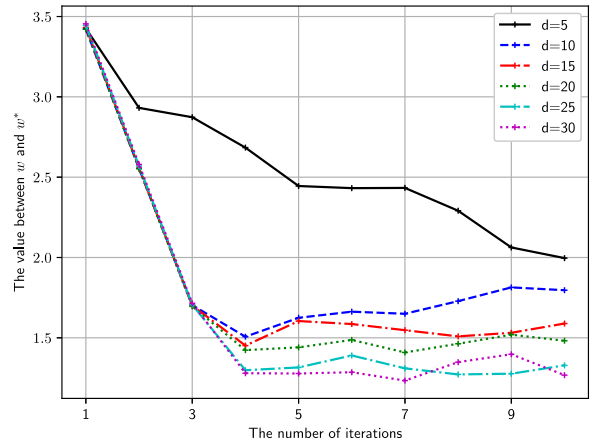
(c) The RMSE between y_{ij} and y_{ij}^*



(d) The RMSE between F and F^*



(e) The RMSE between w_0 and w_0^*



(f) The RMSE between w and w^*

Fig. 5. Experimental results when d varied.

estimation is increased.

From the above discussion, our proposed model can estimate latent structure and clarify the effectiveness of the proposed model.

5. Simulation experiment using real data

To clarify the effectiveness of the proposed model for real data, we conduct the simulation experiment using UCI machine learning repository [21].

5.1 Simulation setting

We use a wine data set in this experiment. This data set presents the results of chemical analysis of wines from three different grape varieties grown in the same region in Italy. The number of data is $n = 178$, the number of input data dimension is $d = 13$, and each data belongs to one of three different categories¹, and the feature names of the wine data set are shown by Table V.

Here, we generate y_{ij} which is a binary relationship between the i -th and the j -th objects. More precisely, it takes 1 if both objects belong to the same category and 0 otherwise. In addition, we normalize input data $\mathbf{X}$ whose mean is $\boldsymbol{\mu}_x = \mathbf{0}$, and set the dimension k of latent vector dimension to be $k = 6$ based on the BIC, and $L = 10$.

Let the notation “ $\hat{\cdot}$ ” denote the estimated result by the proposed model and we calculate the RMSEs $\hat{\mathbf{Y}}$ and $\mathbf{Y}$, and $\hat{\mathbf{X}}$ and $\mathbf{X}$. From the assumption, $\hat{\mathbf{X}}$ is calculated by $\hat{\mathbf{X}} = (\hat{\mathbf{x}}_1, \hat{\mathbf{x}}_2, \dots, \hat{\mathbf{x}}_n)$, where $\hat{\mathbf{x}}_i = \hat{\mathbf{F}}\hat{\mathbf{z}}_i$.

Here, from the formulation of this model, the latent vector $\mathbf{z}_i$ is a low dimensional representation of input data $\mathbf{x}_i$. Moreover, to analyze columns of the projection matrix $\hat{\mathbf{F}}$, the latent factor of $\mathbf{X}$ can be extracted like the Factor analysis [22]. Therefore, latent structure and the characteristics of latent variables can be interpreted to analyze those outputs.

Furthermore, focusing on y_{ij} that is calculated based on the inner product between $\mathbf{z}_i$ and $\mathbf{z}_j$, the effect of each factor for y_{ij} can be extracted. Therefore, we show the maximum and minimum inner products of $\mathbf{z}_i$ and $\mathbf{z}_j$, and we also show, from those vector elements, what factor affects the relational data y_{ij} . To confirm which elements have a significant impact on a relation y_{ij} , we calculate the products of each element of the i -th and the j -th latent vectors.

5.2 Simulation results

From the results of experiment, the RMSE of $\mathbf{X}$ is 0.7924, the RMSE of $\mathbf{Y}$ is 0.1415, and the values of $\hat{w}_0$ and $\hat{w}$ are -1.1820 and 1.7221 , respectively.

In addition, Fig. 6 shows a heat map of $\hat{\mathbf{F}}$, and Table VI shows its values.

From Fig. 6 and Table VI, it is possible to confirm the factors which affect the input data $\mathbf{X}$. For

Table V. The attribute information of wine data set.

Alcohol	Alkalinity of ash	Flavonoids	Color intensity	Proline
Malic acid	Magnesium	Nonflavonoid phenols	Hue	
Ash	Total phenols	Proanthocyanidins	OD280/OD315 of diluted wines	

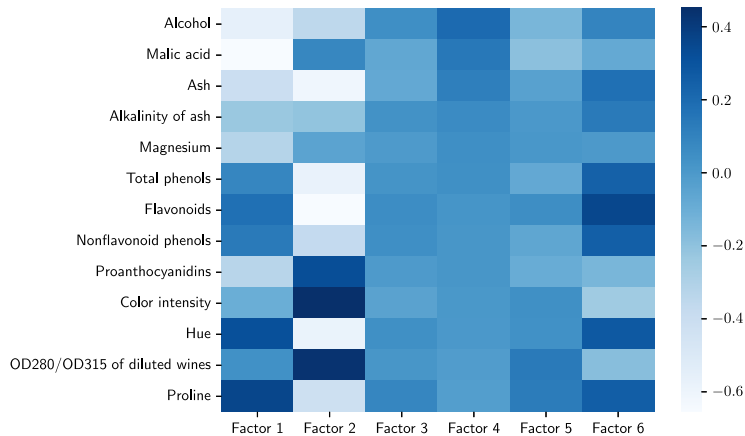


Fig. 6. The heat map of $\hat{\mathbf{F}}$.

¹The number of data in each category is 59 for category 1, 71 for category 2, and 48 for category 3.

Table VI. The value of $\hat{F}$.

	Factor 1	Factor 2	Factor 3	Factor 4	Factor 5	Factor 6
Alcohol	-0.5586	-0.3494	0.0471	0.2023	-0.1403	0.0918
Malic acid	-0.6489	0.0815	-0.0685	0.1464	-0.1907	-0.0792
Ash	-0.4083	-0.6114	-0.0742	0.1152	-0.0399	0.1778
Alkalinity of ash	-0.2294	-0.2100	0.0323	0.0628	0.0029	0.1359
Magnesium	-0.3222	-0.0501	-0.0065	0.0463	0.0124	-0.0026
Total phenols	0.0853	-0.5739	0.0250	0.0422	-0.0754	0.2413
Flavonoids	0.1747	-0.6548	0.0542	0.0191	0.0518	0.3514
Nonflavonoid phenols	0.1387	-0.3704	0.0478	0.0168	-0.0631	0.2484
Proanthocyanidins	-0.3291	0.3209	-0.0052	0.0158	-0.0923	-0.1389
Color intensity	-0.1022	0.4496	-0.0425	0.0075	0.0403	-0.2487
Hue	0.3139	-0.5774	0.0416	0.0012	0.0387	0.2739
OD280/OD315 of diluted wines	0.0363	0.4347	0.0142	-0.0159	0.1349	-0.1841
Proline	0.3563	-0.4151	0.0855	-0.0240	0.1300	0.2535

Table VII. The maximum inner product between two latent vectors.

	1	2	3	4	5	6	inner product	$y_{13,23}$
z_{13}	-0.2667	-0.4459	-2.9725	-2.0502	-0.5173	-2.7257	9.4275	1
z_{23}	-0.6034	0.1189	-1.3969	-1.4387	1.4688	-1.0922		
product	0.1609	-0.0530	4.1525	2.9497	-0.7597	2.9771		

Table VIII. The minimum inner product between two latent vectors.

	1	2	3	4	5	6	inner product	$y_{105,178}$
z_{105}	-1.6500	-0.6792	1.8560	-4.6908	0.4275	2.8624	-19.2603	0
z_{178}	0.2512	0.4750	1.5350	3.5430	0.7973	-1.7794		
product	-0.4145	-0.3227	2.8489	-16.6195	0.3409	-5.0934		

example, the Proline, Hue and Flavonoids have a positive effect, and Magic acid and Alcohol have a negative effect on constructing factor 1. Thus, using the proposed model, it is possible to analyze the latent structure of input data confirming the elements of estimated $\hat{F}$.

Table VII and VIII show the maximum/minimum inner products of the pair of latent vectors and their values. From Table VII, there is a relation between the 13-th and the 23-th objects and factors 3, 4 and 6 have a strong impact on the generation of a relationship. On the contrary, Table VIII shows why the 105-th and the 178-th objects do not have a relation. That is, there is an opposite tendency with respect to factor 4, therefore it seems to have a negative impact on a relation between the 105-th and the 178-th objects.

6. Conclusions and future works

In this paper, we proposed a new probabilistic latent structural model for relational data and parameter estimation method using Laplace approximation and the Monte Carlo EM algorithm under the assumption that the relational data is binary.

Simulation results also showed the effectiveness of our proposed model when appropriately estimating the true values of parameters. In addition, we showed that a suitable number of latent dimensions can be decided using the BIC. On the other hand, there was a possibility that our proposed model might have local minima depending on the initial values. To avoid this situation, it is required to derive an appropriate initial value, which would be one of our future works.

Future studies include investigating the effectiveness of our proposed model for large-scale real data, such as DNA storage media that can store huge amounts of data for ultra long lifespan (*e.g.* [23]). In addition, we aim to apply metaheuristics to optimize the Q function and derive an efficient update algorithm and a way to decide an appropriate number of latent dimensions.

Acknowledgments

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Appendix

A. Derivation of the posterior distribution of $\mathbf{z}_i$

To derive the posterior distribution of $\mathbf{z}_i$ using Laplace approximation, it needs to differentiate Eq. (20) and maximize that to derive $\boldsymbol{\eta}_i$. Here, as mentioned above, $\boldsymbol{\eta}_i$ does not have a closed-form solution, we use the Newton method to optimize $\boldsymbol{\eta}_i$. Moreover, to derive second derivative of Eq. (20) using $\boldsymbol{\eta}_i$, the A_i is obtained.

Here, to differentiate the Eq. (20) with respect to $\mathbf{z}_i$, following equation is obtained.

$$\begin{aligned} \frac{\partial \ln f(\mathbf{Z}|\mathbf{X}, \mathbf{Y}, \boldsymbol{\theta})}{\partial \mathbf{z}_i} = & \frac{\partial}{\partial \mathbf{z}_i} \sum_{i=1}^n \left(\ln \frac{1}{(2\pi)^{\frac{d}{2}} |\boldsymbol{\Sigma}|^{\frac{1}{2}}} - \frac{1}{2} (\mathbf{x}_i - F\mathbf{z}_i - \boldsymbol{\mu}_x)^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - F\mathbf{z}_i - \boldsymbol{\mu}_x) \right. \\ & \left. + \ln \frac{1}{(2\pi)^{\frac{k}{2}} |\sigma_z^2 I|^{\frac{1}{2}}} - \frac{1}{2\sigma_z^2} \|\mathbf{z}_i\|^2 + \frac{1}{2} \sum_{j \neq i} \{y_{ij} \ln s_{ij} + (1 - y_{ij}) \ln(1 - s_{ij})\} \right). \end{aligned} \quad (\text{A-1})$$

Since the fifth term of the right-hand side of Eq. (A-1) depends on $\mathbf{z}_i$, it is defined by $E(\mathbf{Z})$. By differentiating $E(\mathbf{Z})$ with respect to $\mathbf{z}_i$, we have

$$\frac{\partial E(\mathbf{Z})}{\partial \mathbf{z}_i} = \frac{\partial E(\mathbf{Z})}{\partial s_{ij}} \frac{\partial s_{ij}}{\partial \mathbf{z}_i}, \quad (\text{A-2})$$

and $\frac{\partial E(\mathbf{Z})}{\partial s_{ij}}, \frac{\partial s_{ij}}{\partial \mathbf{z}_i}$ can be calculated by

$$\frac{\partial E(\mathbf{Z})}{\partial s_{ij}} = \frac{1}{2} \sum_{j \neq i} \frac{y_{ij} - s_{ij}}{s_{ij}(1 - s_{ij})}, \quad (\text{A-3})$$

$$\frac{\partial s_{ij}}{\partial \mathbf{z}_i} = s_{ij}(1 - s_{ij})w\mathbf{z}_j. \quad (\text{A-4})$$

From the above equations,

$$\frac{\partial E(\mathbf{Z})}{\partial \mathbf{z}_i} = \frac{1}{2} \sum_{j \neq i} (y_{ij} - s_{ij})w\mathbf{z}_j, \quad (\text{A-5})$$

is obtained. Finally, to substitute the other terms that are differentiated by $\mathbf{z}_i$, Eq. (23) is obtained. $\square$

On the other hand, in case using the Laplace approximation, the inverse of the covariance matrix of approximated multi-dimensional Gaussian distribution A_i^{-1} is derived by $A_i^{-1} = -\nabla \nabla \ln f(\mathbf{Z}|\mathbf{X}, \mathbf{Y}, \boldsymbol{\theta})|_{\mathbf{z}_i=\boldsymbol{\eta}_i}$.

It can be derived by differentiating Eq. (23) by $\mathbf{z}_i$ as follows:

$$\frac{\partial^2 \ln f(\mathbf{Z}|\mathbf{X}, \mathbf{Y}, \boldsymbol{\theta})}{\partial \mathbf{z}_i \partial \mathbf{z}_i^\top} = \left(\frac{1}{\sigma_z^2} I + F^\top \boldsymbol{\Sigma}^{-1} F + \frac{1}{2\sigma_y^2} \sum_{j \neq i} s_{ij}(1 - s_{ij})w^2 \mathbf{z}_j \mathbf{z}_j^\top \right)^{-1}. \quad (\text{A-6})$$

$\square$

B. Derivation of the parameters $\boldsymbol{\theta}$

The parameter estimation of our proposed model is implemented to maximize the approximated Q function. In other words, it can be obtained by differentiating Eq. (18) for each parameter.

Here, writing down each term in Eq. (18), we obtain

$$\begin{aligned} \ln p\left(\mathbf{x}_i | \mathbf{z}_i^{(l)}, F, \mathbf{\Sigma}, \boldsymbol{\mu}_x\right) &= -\frac{d}{2} \ln 2\pi - \frac{1}{2} \ln |\mathbf{\Sigma}| \\ &\quad - \frac{1}{2} \left(\mathbf{x}_i - F \mathbf{z}_i^{(l)} - \boldsymbol{\mu}_x\right)^\top \mathbf{\Sigma}^{-1} \left(\mathbf{x}_i - F \mathbf{z}_i^{(l)} - \boldsymbol{\mu}_x\right), \end{aligned} \quad (\text{B-1})$$

$$\ln p\left(y_{ij} | \mathbf{z}_i^{(l)}, \mathbf{z}_j^{(l)}, w_0, w\right) = y_{ij} \ln s_{ij} + (1 - y_{ij}) \ln(1 - s_{ij}), \quad (\text{B-2})$$

$$\ln p\left(\mathbf{z}_i^{(l)}, \sigma_z^2\right) = -\frac{k}{2} \ln 2\pi - \frac{1}{2} \ln |\sigma_z^2 I| - \frac{1}{2} \left\| \mathbf{z}_i^{(l)} \right\|^2. \quad (\text{B-3})$$

The parameters of $F, \boldsymbol{\mu}_x, \mathbf{\Sigma}, \sigma_z^2$ have analytical solutions, therefore, the updating formula can be obtained by differentiating Eqs. (B-1)–(B-3) corresponding parameters.

In case of F , differentiating Eq. (B-1) with respect to F , we have

$$\frac{\partial \sum_{l=1}^L \sum_{i=1}^n \ln p\left(\mathbf{x}_i | \mathbf{z}_i^{(l)}, F, \mathbf{\Sigma}, \boldsymbol{\mu}_x\right)}{\partial F} = \frac{1}{L} \sum_{l=1}^L \sum_{i=1}^n \mathbf{\Sigma}^{-1} \left(\mathbf{x}_i - F \mathbf{z}_i^{(l)} - \boldsymbol{\mu}_x\right) \mathbf{z}_i^{(l)\top}. \quad (\text{B-4})$$

Setting Eq. (B-4) to be $\mathbf{0}$, and to solve this for F ,

$$\sum_{l=1}^L \sum_{i=1}^n F \mathbf{z}_i^{(l)} F \mathbf{z}_i^{(l)\top} = \sum_{l=1}^L \sum_{i=1}^n \mathbf{\Sigma}^{-1} \left(\mathbf{x}_i - \boldsymbol{\mu}_x\right) \mathbf{z}_i^{(l)\top}, \quad (\text{B-5})$$

is obtained. Then updating formula is obtained as follows:

$$F = \sum_{l=1}^L \sum_{i=1}^n \left(\mathbf{x}_i - \boldsymbol{\mu}_x\right) \mathbf{z}_i^{(l)\top} \left(\mathbf{z}_i^{(l)} \mathbf{z}_i^{(l)\top}\right)^{-1}. \quad (\text{B-6})$$

Similarly, for $\boldsymbol{\mu}_x$, by differentiating Eq. (B-1) and setting to $\mathbf{0}$ then following formula is obtained.

$$\sum_{l=1}^L \sum_{i=1}^n \boldsymbol{\mu}_x = \sum_{l=1}^L \sum_{i=1}^n \left(\mathbf{x}_i - F \mathbf{z}_i^{(l)}\right). \quad (\text{B-7})$$

Consequently,

$$\boldsymbol{\mu}_x = \frac{1}{Ln} \sum_{l=1}^L \sum_{i=1}^n \left(\mathbf{x}_i - F \mathbf{z}_i^{(l)}\right), \quad (\text{B-8})$$

is obtained.

The updating formula of $\mathbf{\Sigma}$ is obtained by differentiating the Eq. (B-1). Then we get

$$\begin{aligned} &\frac{\partial \sum_{l=1}^L \sum_{i=1}^n \ln p\left(\mathbf{x}_i | \mathbf{z}_i^{(l)}, F, \mathbf{\Sigma}, \boldsymbol{\mu}_x\right)}{\partial \mathbf{\Sigma}} \\ &= \frac{1}{L} \sum_{l=1}^L \sum_{i=1}^n \left\{ \frac{1}{2} \mathbf{\Sigma}^{-1} + \frac{1}{2} \left(\mathbf{\Sigma}^{-1} \left(\mathbf{x}_i - F \mathbf{z}_i^{(l)} - \boldsymbol{\mu}_x\right) \left(\mathbf{x}_i - F \mathbf{z}_i^{(l)} - \boldsymbol{\mu}_x\right)^\top \mathbf{\Sigma}^{-1} \right) \right\}. \end{aligned} \quad (\text{B-9})$$

Here, setting Eq. (B-9) to be $\mathbf{0}$ and since $\mathbf{\Sigma}$ is a diagonal matrix, then

$$\mathbf{\Sigma} = \text{diag} \left\{ \frac{1}{Ln} \sum_{l=1}^L \sum_{i=1}^n \left(\mathbf{x}_i - F \mathbf{z}_i^{(l)} - \boldsymbol{\mu}_x\right) \left(\mathbf{x}_i - F \mathbf{z}_i^{(l)} - \boldsymbol{\mu}_x\right)^\top \right\}, \quad (\text{B-10})$$

is obtained.

Finally, for the σ_z^2 update, to differentiate the Eq. (B-3), and set to 0, following formula is obtained.

$$\frac{\ln p\left(\mathbf{z}_i^{(l)}, \sigma_z^2\right)}{\partial \sigma_z^2} = \frac{1}{L} \sum_{l=1}^L \sum_{i=1}^n \left(-\frac{k}{2\sigma_z^2} + \frac{1}{2(\sigma_z^2)^2} \left\| \mathbf{z}_i^{(l)} \right\|^2 \right) = 0. \quad (\text{B-11})$$

Then,

$$\sum_{l=1}^L \sum_{i=1}^n \sigma_z^2 = \frac{1}{k} \sum_{l=1}^L \sum_{i=1}^n \left\| \mathbf{z}_i^{(l)} \right\|^2, \quad (\text{B-12})$$

is obtained and updating formula of σ_z^2 is derived as

$$\sigma_z^2 = \frac{1}{Lnk} \sum_{l=1}^L \sum_{i=1}^n \left\| \mathbf{z}_i^{(l)} \right\|^2. \quad (\text{B-13})$$

On the other hand, neither w_0 nor w has a closed-form solution. Therefore, it needs to update those parameters using iterative methods. In this case, we use the Adam which is a gradient-based method to optimize those.

Here, to differentiate the Eq. (B-2) by each parameter,

$$\frac{\partial \ln p \left(y_{ij} | \mathbf{z}_i^{(l)}, \mathbf{z}_j^{(l)}, w_0, w \right)}{\partial w_0} = \frac{1}{L} \sum_{l=1}^L \sum_{i=1}^n \sum_{j \neq i}^n (y_{ij} - s_{ij}), \quad (\text{B-14})$$

$$\frac{\partial \ln p \left(y_{ij} | \mathbf{z}_i^{(l)}, \mathbf{z}_j^{(l)}, w_0, w \right)}{\partial w} = \frac{1}{L} \sum_{l=1}^L \sum_{i=1}^n \sum_{j \neq i}^n (y_{ij} - s_{ij}) \mathbf{z}_i^\top \mathbf{z}_j, \quad (\text{B-15})$$

is obtained respectively. Using those gradients, the parameter update can be implemented.

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